Experiment 7: Lappin and Leass Algorithm for Pronoun Resolution

Aim

To implement the **Lappin and Leass algorithm** for resolving pronouns and identifying their correct antecedents in a sentence.

Theory

Pronoun Resolution is a task in **Natural Language Processing (NLP)** that involves finding the noun (antecedent) to which a pronoun refers in a sentence.

Example:

- *"Riya dropped the glass because she was careless."*
  → "she" refers to Riya

Lappin and Leass Algorithm

The Lappin and Leass algorithm is a **rule-based approach** for pronoun resolution. It assigns scores to possible antecedents and selects the most suitable one.

Working Principle

- **Candidate Identification:**
  Identify possible antecedents (nouns) before the pronoun.
- **Salience Scoring:**
  Assign scores based on factors like:
  - Recency (recent nouns get higher score)
  - Grammatical role (subject > object)
  - Repetition and emphasis
- **Filtering:**
  Remove candidates that do not agree in gender, number, or syntax.
- **Selection:**
  Choose the candidate with the highest score as the correct antecedent.

Conclusion

The Lappin and Leass algorithm is an effective rule-based method for pronoun resolution. It uses syntactic and contextual features to determine the correct antecedent. Although it is simple and does not require training data, its performance is limited compared to modern machine learning approaches. However, it remains useful for understanding the basics of pronoun resolution.

EXPT 8 IS CASE STUDY SHARED ON WHATSAPP